YANG MA

CP3, UCLouvain ⋄ Chemin du Cyclotron 2, 1348 Louvain-la-Neuve, Belgium

yang.ma@uclouvain.be https://yangphy.github.io

PROFESSIONAL PREPARATION

Ph.D. in Physics August 2022
University of Pittsburgh, PA, U.S.A. Advisor: Tao Han

M.S. in Physics June 2016
Chongqing University, Chongqing, China

SKILLS

Programming Language: C/C++, Fortran, Python, Shell script

Handy Programs: Mathematica, Matlab, LaTeX, Excel, Powerpoint, Linux (OS)

HEP Packages: MadGraph5_aMC@NLO, Pythia, FeynRules, FeynArts, FeynCalc, FormCalc, Whizard, ManeParse

POSITION HELD

F.R.S.-FNRS Postdoctoral Fellow (Chargé de recherches) October 2024 - present
Center for Cosmology, Particle Physics and Phenomenology (CP3)
Université catholique de Louvain (UCLouvain), Belgium
Individual research fellowship awarded by the Belgian National Fund for Scientific Research (F.R.S.-FNRS).

International Postdoctoral Fellow September 2022 - September 2024
INFN Sezione di Bologna, Italy
Awarded through a nationwide competition (1 of only 16 positions across all INFN divisions in Italy).

Graduate Teaching Fellow May 2022 - July 2022
Dept. Physics & Astronomy, University of Pittsburgh

Graduate Research Fellow January 2022 - April 2022
Dept. Physics & Astronomy, University of Pittsburgh

Arts & Sciences Pre-Doctoral Fellow September 2020 - December 2021
Kenneth P. Dietrich School of Arts & Sciences, University of Pittsburgh

Graduate Research Assistant January 2020 - August 2020
Dept. Physics & Astronomy, University of Pittsburgh

Graduate Teaching Assistant August 2016 - January 2020
Dept. Physics & Astronomy, University of Pittsburgh

AWARDS AND HONORS

Outstanding Reviewer Awards 2022 March 2023
Journal of Physics Communications, IOP Publishing

DPF Student Travel Award April 2022
APS Division of Particles and Fields (DPF)

Thomas-Lain Scholarship April 2021
Dept. Physics & Astronomy, University of Pittsburgh

FGSA Award for Excellence in Graduate Research February 2021
American Physical Society (APS)

Arts & Sciences Graduate Fellowship September 2020
Kenneth P. Dietrich School of Arts & Sciences, University of Pittsburgh

Pitt Physics and Astronomy China Initiative (PACI) Scholarship September 2016
Dept. Physics & Astronomy, University of Pittsburgh

MEMBERSHIP IN PROFESSIONAL SOCIETIES

American Physical Society (APS) member	March 2021 - present
International Organization of Chinese Physics & Astrophysics member	September 2021 - present
American Association for the Advancement of Science (AAAS) member	September 2021 - present
International Muon collider collaboration (IMCC) member	July 2022 - present

REFEREE SERVICE

- Chinese Physics C (*CPC*) ×3
- Chinese Physics Letters (*Chin. Phys. Lett.*) ×1
- The European Physical Journal C (*Eur. Phys. J. C*) ×1
- Machine Learning: Science and Technology (*MLST*) ×1
- Nuclear Physics B (*Nucl. Phys. B*) ×1
- Journal of High Energy Physics (*JHEP*) ×1
- Journal of Physics Communications (*J. Phys. Commun.*) ×1
- Journal of Physics G: Nuclear and Particle Physics (*J. Phys. G*) ×2
- Physics Letters B (*Phys. Lett. B*) ×1
- Physical Review D (*Phys. Rev. D*) ×1
- Physical Review Letters (*Phys. Rev. Lett.*) ×2

TEACHING EXPERIENCE

Graduate Teaching Fellow (instructor) at the University of Pittsburgh

- PHYS 0174 - *Basic Physics, Science and Engineering 1*, Summer 2022
  Covers Mechanics and Wave

Graduate Teaching Assistant at the University of Pittsburgh

- PHYS 0219 - *Basic Laboratory Physics for Science and Engineering*
  Fall 2016, Spring 2017, and Fall 2018
- PHYS 0212 - *Introduction to Laboratory Physics*
  Fall 2017, Spring 2018, and Summer 2018

- PHYS 0110 - *Introduction to Physics 1*, Summer 2018
  Covers Mechanics, Heat and Thermodynamics, and Waves
- PHYS 0111 - *Introduction to Physics 2*, Summer 2017
  Covers Thermodynamics, Electromagnetism, Optics, Special Relativity, and Quantum Physics
- PHYS 0175 - *Basic Physics, Science and Engineering 2*, Spring 2019 and Summer 2021
  Covers Electromagnetism, Elementary Quantum Mechanics, and Atomic Structure

Graduate Teaching Assistant at Chongqing University

- College Physics I - *Classical Mechanics and Electromagnetism*, Spring 2014
- College Physics II - *Thermodynamics, Optics and Special Relativity*, Fall 2013

MENTORING EXPERIENCE

- Dept. Physics & Astronomy Graduate Student Mentor — August 2021 - May 2022
  Mentor three first year graduate students
- Assist to guide one visiting graduate student (*Xiaoze Tan*) — December 2019 - December 2020
  (*JHEP* **08** (2022) 073 [2202.08273])
- Assist to guide one visiting undergraduate student — June 2019 - August 2019

PUBLICATIONS IN REFEREED JOURNALS

1. P. Bredt, T. Banno, M. Höfer, S. Iguro, W. Kilian, Y. Ma et al., *Chasing the Two-Higgs-Doublet Model via Electroweak Corrections at e+e- Colliders*, *Phys. Rev. Lett.* **136** (2026) 081801 [2509.05421].
2. F. Kling, Y. Ma, K. Mękała, J. Reuter and Z. Tabrizi, *Non-Standard Neutrino Interactions at a Muon Collider Neutrino Detector*, 2508.00761.
3. S.-s. Bao, Y. Ma, Y. Wu, K. Xie and H. Zhang, *Light Axion-Like Particles at Future Lepton Colliders*, *JHEP* **10** (2025) 122 [2505.10023].
4. Y. Ma, D. Pagani and M. Zaro, *EW corrections and heavy boson radiation at a high-energy muon collider*, *Phys. Rev. D* **111** (2025) 053002 [2409.09129].
5. Y. Ma and Z. Wang, *New probe of dark matter-baryon interactions in compact stellar systems*, *Phys. Rev. D* **111** (2025) L061302 [2408.01818].
6. E. Celada, T. Han, W. Kilian, N. Kreher, Y. Ma, F. Maltoni et al., *Probing Higgs-muon interactions at a multi-TeV muon collider*, *JHEP* **08** (2024) 021 [2312.13082].
7. C. Accettura et al., *Towards a muon collider*, *Eur. Phys. J. C* **83** (2023) 864 [2303.08533].
8. T. Han, A. K. Leibovich, Y. Ma and X.-Z. Tan, *Higgs boson decay to charmonia via c-quark fragmentation*, *JHEP* **08** (2022) 073 [2202.08273].
9. T. Han, W. Kilian, N. Kreher, Y. Ma, J. Reuter, T. Striegl et al., *Precision Test of the Muon-Higgs Coupling at a High-energy Muon Collider*, *JHEP* **12** (2021) 162 [2108.05362].
10. D. Buarque et al., *Vector Boson Scattering Processes: Status and Prospects*, *Rev. Phys.* **8** (2022) 100071 [2106.01393].
11. T. Han, Y. Ma and K. Xie, *Quark and gluon contents of a lepton at high energies*, *JHEP* **02** (2022) 154 [2103.09844].

12. T. Han, Y. Ma and K. Xie, *High energy leptonic collisions and electroweak parton distribution functions*, *Phys. Rev. D* **103** (2021) L031301 [2007.14300].

13. Z. Sun and Y. Ma, *Inclusive productions of* $\Upsilon(1S, 2S, 3S)$ *and* $\chi_b(1P, 2P, 3P)$ *via the Higgs boson decay*, *Phys. Rev. D* **100** (2019) 094019 [1909.08548].

14. Z. Sun, X.-G. Wu, Y. Ma and S. J. Brodsky, *Exclusive production of* $J/\psi + \eta_c$ *at the B factories Belle and Babar using the principle of maximum conformality*, *Phys. Rev. D* **98** (2018) 094001 [1807.04503].

15. Y. Ma and X.-G. Wu, *Renormalization scheme dependence of high-order perturbative QCD predictions*, *Phys. Rev. D* **97** (2018) 036024 [1707.09886].

16. J.-M. Shen, X.-G. Wu, Y. Ma and S. J. Brodsky, *The Generalized Scheme-Independent Crewther Relation in QCD*, *Phys. Lett. B* **770** (2017) 494 [1611.07249].

17. H.-Y. Bi, X.-G. Wu, Y. Ma, H.-H. Ma, S. J. Brodsky and M. Mojaza, *Degeneracy Relations in QCD and the Equivalence of Two Systematic All-Orders Methods for Setting the Renormalization Scale*, *Phys. Lett. B* **748** (2015) 13 [1505.04958].

18. H.-H. Ma, X.-G. Wu, Y. Ma, S. J. Brodsky and M. Mojaza, *Setting the renormalization scale in perturbative QCD: Comparisons of the principle of maximum conformality with the sequential extended Brodsky-Lepage-Mackenzie approach*, *Phys. Rev. D* **91** (2015) 094028 [1504.01260].

19. Y. Ma, X.-G. Wu, H.-H. Ma and H.-Y. Han, *General Properties on Applying the Principle of Minimum Sensitivity to High-order Perturbative QCD Predictions*, *Phys. Rev. D* **91** (2015) 034006 [1412.8514].

20. H.-B. Fu, X.-G. Wu and Y. Ma, $B \to K^*$ *Transition Form Factors and the Semi-leptonic Decay* $B \to K^*\mu^+\mu^-$, *J. Phys. G* **43** (2016) 015002 [1411.6423].

21. H.-B. Fu, X.-G. Wu, H.-Y. Han, Y. Ma and H.-Y. Bi, *The* ρ*-meson longitudinal leading-twist distribution amplitude*, *Phys. Lett. B* **738** (2014) 228 [1409.3053].

22. G. Chen, X.-G. Wu, Z. Sun, Y. Ma and H.-B. Fu, *Photoproduction of doubly heavy baryon at the ILC*, *JHEP* **12** (2014) 018 [1408.4615].

23. H.-B. Fu, X.-G. Wu, H.-Y. Han and Y. Ma, $B \to \rho$ *transition form factors and the* ρ*-meson transverse leading-twist distribution amplitude*, *J. Phys. G* **42** (2015) 055002 [1406.3892].

24. X.-G. Wu, Y. Ma, S.-Q. Wang, H.-B. Fu, H.-H. Ma, S. J. Brodsky et al., *Renormalization Group Invariance and Optimal QCD Renormalization Scale-Setting*, *Rept. Prog. Phys.* **78** (2015) 126201 [1405.3196].

25. S.-Q. Wang, X.-G. Wu, J.-M. Shen, H.-Y. Han and Y. Ma, *QCD improved electroweak parameter* ρ, *Phys. Rev. D* **89** (2014) 116001 [1402.0975].

26. Z. Sun, X.-G. Wu, G. Chen, Y. Ma, H.-H. Ma and H.-Y. Bi, *Bottomonium production associated with a photon at a high luminosity* e^+e^- *collider with the one-loop QCD correction*, *Phys. Rev. D* **89** (2014) 074035 [1401.2735].

27. H.-B. Fu, X.-G. Wu, H.-Y. Han, Y. Ma and T. Zhong, $|V_{cb}|$ *from the semileptonic decay* $B \to D\ell\bar{\nu}_\ell$ *and the properties of the D meson distribution amplitude*, *Nucl. Phys. B* **884** (2014) 172 [1309.5723].

OTHER PUBLICATIONS

1. MuCoL collaboration, *MuCol Milestone Report No. 7: Consolidated Parameters*, 2510.27437.
2. CEPC Study Group collaboration, *CEPC Technical Design Report - Reference Detector*, 2510.05260.
3. J. Altmann et al., *ECFA Higgs, electroweak, and top Factory Study*, vol. 5/2025 of *CERN Yellow Reports: Monographs*. 6, 2025, 10.23731/CYRM-2025-005, [2506.15390].
4. International Muon Collider collaboration, *The Muon Collider*, 2504.21417.
5. MuCoL collaboration, *MuCol Milestone Report No. 5: Preliminary Parameters*, 2411.02966.
6. Y. Ma, E. Celada, T. Han, W. Kilian, N. Kreher, F. Maltoni et al., *Higgs-muon interactions at a multi-TeV muon collider*, *PoS* **ICHEP2024** (2024) 092 [2410.06991].
7. International Muon Collider collaboration, *Interim report for the International Muon Collider Collaboration (IMCC)*, 2407.12450.
8. CEPC Study Group collaboration, *CEPC Technical Design Report: Accelerator*, *Radiat. Detect. Technol. Methods* **8** (2024) 1 [2312.14363].
9. J. Reuter, T. Han, W. Kilian, N. Kreher, Y. Ma, T. Striegl et al., *Precision test of the muon-Higgs coupling at a high-energy muon collider*, *PoS* **ICHEP2022** (2022) 1239 [2212.01323].
10. T. Han, A. K. Leibovich, Y. Ma and X.-Z. Tan, *Higgs decay to charmonia and the charm-quark Yukawa coupling*, *PoS* **ICHEP2022** (2022) 517 [2211.10727].
11. K. M. Black et al., *Muon Collider Forum report*, *JINST* **19** (2024) T02015 [2209.01318].
12. T. Han, Y. Ma and K. Xie, *Electroweak fragmentation at high energies: A Snowmass White Paper*, in *Snowmass 2021*, 3, 2022, 2203.11129.
13. J. M. Campbell et al., *Event Generators for High-Energy Physics Experiments*, *SciPost Phys.* **16** (2024) 130 [2203.11110].
14. C. Aime et al., *Muon Collider Physics Summary*, 2203.07256.
15. Muon Collider collaboration, *The physics case of a 3 TeV muon collider stage*, 2203.07261.
16. ILC International Development Team collaboration, *The International Linear Collider: Report to Snowmass 2021*, 2203.07622.

SEMINARS AND COLLOQUIUM

1. *A New Frontier: Charting the Course for a 10 TeV Muon Collider* — October 2025
   SOTU Seminar, Tata Institute of Fundamental Research, Mumbai, India
2. *Precision physics at the future muon colliders* — May 2025
   Theory Colloquium, DESY, Germany
3. *Probe the Yukawa interactions of the 2nd generation fermions at high-energy colliders* — July 2024
   TDLI/INPAC Joint Theory Seminar, Shanghai Jiao Tong University
4. *Physics opportunities and challenges at future multi-TeV lepton colliders* — December 2023
   HEP Seminar, Nankai University
5. *Physics opportunities and challenges at future multi-TeV lepton colliders* — December 2023
   HEP Seminar, Chongqing University

6. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Central South University

7. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Hunan University

8. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Shandong University

9. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, University of Science and Technology of China (USTC)

10. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Nanjing Normal University

11. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Southeast University

12. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Fudan University

13. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Nanjing University

14. *Physics opportunities and challenges at future multi-TeV lepton colliders* December 2023
HEP Seminar, Tsung-Dao Lee Institute (TDLI), Shanghai Jiao Tong University

15. *Physics opportunities and challenges at future multi-TeV lepton colliders* November 2023
HEP Seminar, Tsinghua University

16. *Physics opportunities and challenges at future multi-TeV lepton colliders* November 2023
HEP Seminar, Peking University

17. *Physics opportunities and challenges at future multi-TeV lepton colliders* November 2023
HEP Seminar, Institute of Theoretical Physics, Chinese Academy of Sciences

18. *Physics opportunities and challenges at future multi-TeV lepton colliders* November 2023
HEP Theory Seminar, IHEP, Chinese Academy of Sciences

19. *Bread and butter physics at future multi-TeV lepton colliders* May 2023
NHETC Theory Seminar, Rutgers University

20. *Measuring the Yukawa couplings: Towards the 2nd generation fermions* May 2023
HEP Lunch Seminar, University of Chicago

21. *Electroweak Tevatron: High-Energy lepton colliders* May 2023
Theoretical Physics Seminar, Fermilab

22. *Bread and butter physics at future multi-TeV lepton colliders* May 2023
LEPP Theory Seminar, Cornell University

23. *Measuring the Yukawa couplings: Towards the 2nd generation fermions* May 2023
HEP Theory Seminar, University at Buffalo

24. *Determine the Yukawa couplings of the second generation fermions at high-energy colliders* January 2023
(Remote) HEP Theory Seminar, Argonne National Laboratory (ANL)

25. *Determine the Yukawa couplings of the second generation fermions at high-energy colliders* December 2022
(Remote) TDLI/INPAC Joint Theory Seminar, Shanghai Jiao Tong University

26. *Determine the Yukawa couplings of the second generation fermions at high-energy colliders* — December 2022
(Remote) Particle Physics Seminar, Chongqing University

27. *Determine the Yukawa couplings of the second generation fermions at high-energy colliders* — December 2022
(Remote) Theoretical Physics Seminar, Shandong University

28. *Phenomenology at high-energy colliders* — November 2022
Bologna HEP Theory Journal Club, INFN Bologna & University of Bologna

29. *The partonic picture and the SM expectation of high-energy lepton colliders* — September 2022
HEP Seminar, University of Notre Dame

30. *The partonic picture and the SM expectation of high-energy lepton colliders* — September 2022
HEP Seminar, University of Wisconsin-Madison

31. *The partonic picture and the SM expectation of high-energy lepton colliders* — September 2022
HEP Special Seminar, University of Michigan

32. *The partonic picture and the SM expectation of high-energy lepton colliders* — September 2022
HEP Seminar, Northwestern University

33. *Higgs decay to charmonia and the charm-quark Yukawa coupling* — September 2022
HEP Seminar, Michigan State University

34. *Higgs decay to charmonia and the charm-quark Yukawa coupling* — September 2022
HEP Seminar, Washington University in St. Louis

35. *The partonic picture and the SM expectation of high-energy lepton colliders* — August 2022
HEP Seminar, University of Minnesota

36. *Higgs decay to J/ψ via c-quark fragmentation* — May 2022
(Remote) Nuclear Physics Seminar, UCLA

37. *Higgs decay to charmonia and the charm quark Yukawa* — March 2022
PITT PACC Group Seminar, University of Pittsburgh

38. *Multi-boson production and the muon Yukawa coupling* — October 2021
(Remote) HEP Journal Club, University of Utah

39. *Multi-boson production and the muon Yukawa coupling* — September 2021
PITT PACC Group Seminar, University of Pittsburgh

40. *Parton contents of a lepton at high energies* — May 2021
(Remote) Particle Theory Seminar, Carleton University

41. *The partonic picture at high-energy lepton colliders* — April 2021
(Remote) SLAC EPP Theory Seminar, SLAC

42. *The partonic picture at high-energy lepton colliders* — April 2021
(Remote) Particle Theory Seminar, Shandong University

43. *Parton contents of a lepton at high energies* — April 2021
(Remote) HEP Seminar, Oklahoma State University

44. *QCD jet production at high energy lepton colliders* — March 2021
(Remote) PITT PACC Group Seminar, University of Pittsburgh

45. *High energy lepton collisions and electroweak PDFs* — October 2020
(Remote) Particle Theory Seminar, Carleton University

46. *High energy lepton collisions and electroweak PDFs* September 2020
(Remote) PITT PACC Group Seminar, University of Pittsburgh

47. *How much do we need polarized PDFs?* October 2019
PITT PACC Group Seminar, University of Pittsburgh

48. *Renormalization scheme uncertainties in high order perturbative QCD results* March 2019
PITT PACC Group Seminar, University of Pittsburgh

CONFERENCE AND WORKSHOP TALKS

1. *Toward precision Higgs at FCC-ee and jet physics at the muon collider* February 2026
MadGraph5_aMC@NLO meeting 2026, Heidelberg, Germany

2. *Precision physics at the future muon collider* August 2025
(Invited) Parallel talk at the 2nd Annual US Muon Collider Meeting, University of Chicago

3. *Higgs-muon interactions at a multi-TeV muon collider* July 2025
GGI workshop: *Exploring the energy frontier with muon beams*, Florence, Italy

4. *Light Axion-Like Particles at Future Lepton Colliders* May 2025
Parallel talk at Pheno 2025, University of Pittsburgh

5. *Probing electroweak NLO corrections and Sudakov logarithms at multi-TeV muon colliders* May 2025
Parallel talk at IMCC and MuCol Annual Meeting 2025, DESY, Germany

6. *Precision physics at the future muon collider* February 2025
MadGraph5_aMC@NLO meeting 2025, CERN

7. *Higgs-muon interactions at a multi-TeV muon collider* November 2024
Parallel talk at Higgs 2024, Uppsala University, Sweden

8. *Charm and bottom Yukawa couplings via quarkonia production at HL-LHC* November 2024
Parallel talk at Higgs 2024, Uppsala University, Sweden

9. *Electroweak radiation picture of the future multi-TeV muon collider* August 2024
Early Career Researchers & Muon Colliders (Online Event)

10. *Higgs-muon interactions at a multi-TeV muon collider* July 2024
Parallel talk at ICHEP 2024, Prague, Czech Republic

11. *Towards a Muon Collider: III. Higgs Physics* June 2024
Frontier Physics Working Month, South China Normal University

12. *Towards a Muon Collider: II. The Partonic Picture* June 2024
Frontier Physics Working Month, Shandong University (Qingdao)

13. *Towards a Muon Collider: I. The General Picture* June 2024
Frontier Physics Working Month, Peking University

14. *Probing Higgs-Muon Interactions at Multi-TeV Collider* March 2024
Parallel talk at IMCC and MuCol Annual Meeting 2024, CERN

15. *Multiple boson production at high-energy muon colliders to probe the Higgs-muon coupling* December 2023
Parallel talk at Higgs 2023, IHEP, Beijing

16. *Higgs decay to quarkonia and the Yukawa couplings* November 2023
Parallel talk at Higgs 2023, IHEP, Beijing

17. *Muon colliders and Weak PDFs* September 2023
MadGraph5_aMC@NLO meeting 2023, Gargnano, Lake Garda, Italy

18. *Muon Yukawa couplings at the high-energy muon collider* May 2023
Parallel talk at Pheno 2023, University of Pittsburgh

19. *Electroweak LHC: High-energy lepton colliders* April 2023
Invited talk at PIKIMO Spring 2023, Ohio State University

20. *EW and QCD physics at the muon collider* December 2022
Parallel talk at Milan Christmas Meeting 2022, Milan, Italy

21. *Higgs decay to charmonia and the charm-quark Yukawa coupling* November 2022
Parallel talk at the Higgs 2022 Conference, Pisa, Italy

22. *EW and QCD physics at the muon collider* October 2022
Parallel talk at Muon Collider Collaboration Meeting 2022, CERN

23. *Higgs decay to charmonia via c-quark fragmentation* September 2022
Invited plenary talk at QWG 2022, GSI Darmstadt, Germany

24. *Higgs decay to charmonia and the charm-quark Yukawa coupling* September 2022
(Remote) Invited talk at the SYSU-PKU Collider Physics forum For Young Scientists

25. *Higgs decay to J/ψ via c-quark fragmentation* July 2022
(Remote) Parallel talk at ICHEP 2022, Bologna, Italy

26. *Higgs decay to J/ψ via c-quark fragmentation* May 2022
Parallel talk at Pheno 2022, University of Pittsburgh

27. *Multi-boson production and the muon Yukawa coupling* April 2022
Contributed talk at APS April Meeting 2022, New York

28. *Multi-boson production and the muon Yukawa coupling* December 2021
PIKIMO 11, University of Pittsburgh

29. *Electroweak parton distributions and fragmentations for high-energy lepton colliders* October 2021
(Remote) Snowmass EF04 Topical Group Community Meeting

30. *Higgs boson decay to J/ψ via c-quark fragmentation* October 2021
(Remote) Parallel talk at Higgs 2021 Conference, Stony Brook University

31. *The partonic picture at high-energy lepton colliders* August 2021
(Remote) Parallel talk at SUSY 2021, Shanghai

32. *QCD jet production at a high energy muon collider* July 2021
(Remote) Parallel talk at EPS-HEP 2021, DESY

33. *Quark and gluon contents of a lepton at high energies* July 2021
(Remote) Parallel talk at the DPF meeting, Florida State University

34. *Quark and gluon contents of a lepton at high energies* May 2021
(Remote) Parallel talk at Pheno 2021, University of Pittsburgh

35. *The partonic picture at high-energy lepton colliders* May 2021
(Remote) Parallel talk at PPC 2021, University of Oklahoma

36. *Electroweak parton distribution functions at a high-energy muon collider* April 2021
(Remote) Contributed talk at APS April Meeting 2021

37. *QCD jet production at a high energy muon collider* March 2021
(Remote) Talk at Muon Collider Physics and Simulation Meeting, CERN

38. *The electroweak parton distribution functions - Necessity and application* June 2020
(Remote) Student talk at Theoretical Advanced Study Institute (TASI 2020)

39. *The electroweak parton distribution functions* May 2020
(Remote) Parallel talk at Pheno 2020, University of Pittsburgh

40. *QCD Scale-setting problem in Future Chinese Collider physics* September 2015
Parallel talk at CEPC-SppC Study Group Meeting, IHEP, Beijing

CONFERENCES AND WORKSHOPS ATTENDED

1. MadGraph5_aMC@NLO meeting 2026, Heidelberg, Germany February 2026
2. Second Annual US Muon Collider Meeting, University of Chicago July 2025
3. GGI workshop: *Exploring the energy frontier with muon beams*, Florence, Italy July 2025
4. Phenomenology Symposium 2025 (Pheno 2025), University of Pittsburgh May 2025
5. IMCC and MuCol Annual Meeting 2025, DESY, Germany May 2025
6. MadGraph5_aMC@NLO meeting 2025, CERN February 2025
7. Higgs 2024, Uppsala University, Sweden November 2024
8. Early Career Researchers & Muon Colliders (Online Event) August 2024
9. The 42nd International Conference on High Energy Physics (ICHEP 2024) July 2024
Prague, Czech Republic
10. Frontier Physics Working Month 2024, Peking University June 2024
11. IMCC and MuCol Annual Meeting 2024, CERN March 2024
12. Higgs 2023, IHEP, Beijing November 2023
13. MadGraph5_aMC@NLO meeting 2023, Gargnano, Lake Garda, Italy September 2023
14. Muon Collider Collaboration Meeting 2023, IJCLab in Orsay, France May 2023
15. Phenomenology Symposium 2023 (Pheno 2023), University of Pittsburgh May 2023
16. PIKIMO Spring 2023, Ohio State University April 2023
17. Milan Christmas Meeting 2022, Milan, Italy December 2022
18. Higgs 2022, Pisa, Italy November 2022
19. Muon Collider Collaboration Meeting 2022, CERN October 2022
20. The 15th International Workshop on Heavy Quarkonium (QWG 2022) September 2022
GSI Darmstadt, Germany
21. SYSU-PKU Collider Physics forum For Young Scientists (remote) September 2022
22. Snowmass Community Summer Study Workshop (Snowmass 2022), Seattle July 2022
23. XLI International Conference on High Energy Physics (ICHEP 2022), Bologna, Italy July 2022
24. LoopFest XX, University of Pittsburgh May 2022
25. Phenomenology Symposium 2022 (Pheno 2022), University of Pittsburgh May 2022

26. APS April Meeting 2022, New York — April 2022
27. PIKIMO 11, University of Pittsburgh (hybrid) — December 2021
28. Higgs 2021 — October 2021
    Stony Brook University & Brookhaven National Laboratory (remote)
29. The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021), Shanghai (remote) — August 2021
30. European Physical Society Conference on High Energy Physics 2021 (EPS-HEP 2021)
    DESY (remote) — July 2021
31. 2021 Meeting of the Division of Particles and Fields of the APS (DPF21) — July 2021
    Florida State University (remote)
32. Phenomenology Symposium 2021 (Pheno 2021) — May 2021
    University of Pittsburgh (remote)
33. XIV International Workshop on Interconnections between Particle Physics and Cosmology (PPC 2021), University of Oklahoma (remote) — May 2021
34. APS April Meeting 2021 (remote) — April 2021
35. Muon Collider Physics and Simulation Meeting (remote) — March 2021
36. PITT PACC Workshop: Muon collider physics — November 2020
    University of Pittsburgh (remote)
37. Phenomenology Symposium 2020 (Pheno 2020) — May 2020
    University of Pittsburgh (remote)
38. Phenomenology Symposium 2020 (Pheno 2019), University of Pittsburgh — May 2019
39. PITT PACC Workshop: BSM circa 2020, University of Pittsburgh — March 2019
40. Phenomenology Symposium 2020 (Pheno 2018), University of Pittsburgh — May 2018
41. Phenomenology Symposium 2020 (Pheno 2017), University of Pittsburgh — May 2017
42. The CEPC-SppC Study Group Meeting — September 2015
    Institute of High Energy Physics (IHEP), Beijing

SUMMER SCHOOLS ATTENDED

1. **CTEQ 2022**, University of Pittsburgh — July 2022
   CTEQ School on QCD and Electroweak Phenomenology
2. **SSI 2021**, SLAC — August 2021
   49th SLAC SUMMER INSTITUTE: The Higgs State Fair
3. **HCPSS 2020**, Fermilab — August 2020
   15th annual Fermilab-CERN Hadron Collider Physics Summer School
4. **TASI 2020**, University of Colorado Boulder — June 2020
   The Obscure Universe: Neutrinos and Other Dark Matters
5. **CTEQ 2019**, University of Pittsburgh — July 2019
   CTEQ School on QCD and Electroweak Phenomenology
6. **CTEQ 2017**, University of Pittsburgh — July 2017
   CTEQ School on QCD and Electroweak Phenomenology